

Proof-of-Concept Extraction of Registered Gravitational Signature Structure

Retrospective Proxy Evidence and a Tensor-Ablation Certification Standard

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Final interim GCS IV formulation — July 2026

Abstract

The first proof-of-concept GCS IV experiments asked whether many-source gravitational readability structure could be extracted at all. Using the declared weak-field proxy $Q_i \propto M_i/r_i^3$, they produced a three-part operational result: subordinate support persisted under strong hierarchy, dense scenes yielded interface-bearing support architecture, and matched source budgets produced non-degenerate responses under changes of spatial arrangement. Those experiments remain historically useful, but their scalar source-assignment rule no longer carries the primary evidentiary burden of the revised GCS program.

The modern construction begins with solved fused scenes. For each declared source i , the source is ablated, the comparison scene is re-solved, and the electric-Weyl tensor is registered to a common support. The primary source-conditioned signature is

$$\Delta_i E_{ab} = E_{ab}^{(\text{full})} - \tilde{E}_{ab}^{(-i)},$$

and the Frobenius amplitude $S_i = \|\Delta_i E\|_{F,\text{ref}}$ is the default direction-agnostic scalarization. Source-context parity surfaces, many-source signature networks, low-margin regions, and fixed-skeleton relational-complex differences are derived from that tensor evidence.

This interim results paper retrofits the original extraction body to the upgraded framework while preserving claim discipline. The original three experiments are retained as *legacy proxy demonstrations*: they establish that an extraction pipeline can recover hierarchy-, density-, and morphology-sensitive support under one transparent weak-field rule. They do not certify the corresponding structures under registered tensor ablation. A fused-scene certification standard is therefore defined, but no result from that program is reported as complete. Its required outputs are full-versus-ablated tensor fields, FN parity and dominance geometry, fixed-skeleton complex differences, and sensitivity to resolution, masks, registration, and scalarization. GCS IV thus remains the results gateway of the series, but its present threshold claim is deliberately staged: operational extraction has been demonstrated in the proxy lane; tensor-ablation certification is the next laboratory result required for promotion.

Keywords: general relativity; electric Weyl tensor; tensor ablation; Frobenius norm; gravitational coherence surface; many-source signature network; proof-of-concept extraction; morphology sensitivity; relational complex; certification.

1. Introduction

The GCS program asks how finite-source gravitational structure can be read inside a shared, fused scene without pretending that general relativity supplies privately owned field parcels. GCS I

defines registered source-context parity through full-versus-ablated solved scenes. GCS II constructs many-source signature networks from registered ablation responses. GCS III asks which faces, seams, margins, and complex differences persist across changes of scale and coarse-graining [1, 2, 3].

GCS IV has a different job. It must show that these objects can be extracted rather than merely defined.

The April 2026 paper approached that task through a deliberately simple scalar proxy,

$$Q_i(x) = k \frac{M_i}{r_i(x)^3}, \quad (1)$$

with pairwise residuals $\Psi_{ij} = Q_i - Q_j$. Under that lane, retained support could be sampled, rendered, perturbed, and compared. The work established an important threshold: many-source comparison rules can generate operationally recoverable morphology rather than only formal zero sets.

The scientific center has since moved. Equation (1) remains an analytic control and historical benchmark, but it cannot represent the primary source-attribution mechanism in a nonlinear fused geometry. The revised paper therefore performs two tasks at once:

1. preserve and accurately classify the original extraction triad;
2. define the registered-ablation experiments required to certify the same triad under the modern tensor-first framework.

No new fused-scene runs are claimed in this revision. That boundary is intentional. Retrofitting the paper body is useful; retrofitting old proxy outputs into tensor evidence would not be.

2. Evidence hierarchy and claim status

The revised GCS IV distinguishes three evidentiary levels.

Table 1: Evidence hierarchy used in this paper.

Level	Meaning	Status here
Definition	A mathematical object or extraction protocol is specified.	Registered tensor signatures, FN amplitudes, parity geometry, and certification requirements are defined.
Legacy demonstration	An earlier declared proxy lane produced recoverable structure.	The three April 2026 experiments remain valid within $Q_i \propto M_i/r_i^3$.
Certified result	The modern fused-scene construction survives declared sensitivity tests.	Not yet reported for the three GCS IV experiment classes.

The distinction prevents two opposite errors. The legacy work should not be discarded merely because the program acquired a stronger instrument. It demonstrated extraction logic, artifact discipline, and morphology-sensitive comparison. But neither should those outputs be relabeled as electric-Weyl ablation results after the fact.

The present paper therefore makes one explicit interim threshold claim:

Operational extraction of many-source relational support has been demonstrated in a transparent weak-field proxy lane. Promotion of the same experimental triad to the revised GCS framework requires registered fused-scene ablation and sensitivity certification.

3. Modern extraction frame

Let $(\Sigma, \gamma_{ab}, K_{ab})$ be a declared full scene containing sources $\{S_i\}_{i=1}^N$. For each source i , construct a comparison scene with i removed and re-solve under the same declared physical and numerical method. Let I_i and P_i provide the declared point and frame registration from the full-scene support to the ablated scene. The registered ablated electric-Weyl tensor is

$$\tilde{E}_{ab}^{(-i)}(x) = P_a^c(x) P_b^d(x) E_{cd}^{(-i)}(I_i(x)). \quad (2)$$

The source-conditioned response is

$$\Delta_i E_{ab}(x) = E_{ab}^{(\text{full})}(x) - \tilde{E}_{ab}^{(-i)}(x). \quad (3)$$

It is the response of the entire solved scene to the presence of i , including altered nonlinear couplings. It is not asserted to be a private field component of the removed source.

Given a declared positive-definite reference metric, the default scalar amplitude is

$$S_i(x) = \|\Delta_i E(x)\|_{F, \text{ref}} = \left(\gamma_{\text{ref}}^{ac} \gamma_{\text{ref}}^{bd} \Delta_i E_{ab} \Delta_i E_{cd} \right)^{1/2}. \quad (4)$$

The registered ablated-scene contextual amplitude is

$$C_i(x) = \|\tilde{E}^{(-i)}(x)\|_{F, \text{ref}}. \quad (5)$$

The source-context GCS is the parity set

$$\Sigma_{\text{GCS}}^{(i)} = \{x : S_i(x) = C_i(x)\}. \quad (6)$$

For many-source comparison, candidate pairwise signature parity satisfies

$$\Sigma_{ij} = \{x : S_i(x) = S_j(x)\}. \quad (7)$$

On a declared active support U_{act} , the retained dominant face is

$$\overline{\mathcal{F}}_{ij} = \{x \in U_{\text{act}} : S_i = S_j \geq S_k \text{ for all } k\}. \quad (8)$$

Strict dominance interiors, higher-order tie strata, masks, near-tie thresholds, and confidence floors follow the conventions of revised GCS II.

A second extraction object compares relational complexes before and after intervention. Let $\mathcal{K}^{(\text{full})}$ and $\mathcal{K}^{(-i)}$ be evaluated on one fixed or explicitly registered datum skeleton. Their ordered difference

$$\Delta \mathcal{K}_i = \left(\mathcal{R}_i^-, \mathcal{R}_i^+, \mathcal{R}_i^{\leftrightarrow} \right) \quad (9)$$

records removed, born, and relabeled strata. Equations (3)–(4) remain the primary continuous evidence; the complex difference is a derived topological fingerprint.

4. The experimental triad

The original paper organized its proof of concept around three failure modes. That design remains strong and is retained unchanged at the level of experimental logic.

Table 2: Preserved GCS IV experimental triad.

Experiment	Diagnostic question	Modern target
Hierarchy	Does a dominant environment erase subordinate relational structure?	Subordinate FN signature, source GCS, dominance cell, and anisotropic compression.
Dense scene	Can many-source competition produce extractable relational architecture?	Retained FN faces, seams, nodes, margins, and fixed-skeleton complex structure.
Matched budget	Does geometry respond to arrangement beyond total mass and source count?	Quantified differences in network, shell, margin, and ablation-fingerprint morphology.

4.1 Experiment I: persistence under hierarchy

The legacy test placed a subordinate source inside a strongly dominant environment. Under the proxy rule, a finite anisotropic local support pocket remained recoverable rather than collapsing into a global winner-takes-all wash. The pocket was compressed and displaced toward the weaker source side of the comparison.

That remains a valid proxy result. Its modern replacement is more precise. A fused scene should be solved, the subordinate source ablated, and the following objects extracted:

- the tensor field $\Delta_i E_{ab}$ and FN amplitude S_i ;
- the source-context parity set $S_i = C_i$;
- the source-conditioned dominance cell within the multi-ablation network;
- directional shell radius, volume, centroid displacement, and anisotropy;
- sensitivity to norm, resolution, masks, and registration convention.

The modern threshold is not that the subordinate source owns a finite pocket. It is that its registered whole-scene response remains structured and locally discriminable under strong hierarchy.

4.2 Experiment II: emergence in dense many-source scenes

The legacy dense-scene test showed that retained proxy support could assemble into interface-bearing morphology without an imposed Voronoi tessellation. Early graphics were conservatively withheld when later rendering and extraction corrections made their presentation uncertain. That artifact discipline remains exemplary.

The modern test should compute one full scene and one registered ablation per source, construct all S_i , and extract:

- strict dominance interiors and closed dominance cells;
- retained pairwise faces on U_{act} ;
- triple seams and higher-order nodes under transversality checks;
- normalized dominance margins and low-margin volume;
- fixed-skeleton label and stratum inventories;
- feature correspondence across resolution and threshold families.

The word *foam* may remain a visual shorthand for the cell-and-seam grammar, but not an ontological claim. The certification target is a reproducible relational complex derived from registered tensor responses.

4.3 Experiment III: matched-budget morphology sensitivity

The legacy matched-budget comparison held a declared source budget fixed while changing spatial arrangement. Smooth, weak-arm, and lopsided ensembles produced non-degenerate proxy support responses. The result demonstrated arrangement sensitivity under the declared branch, not a morphology invariant.

The modern experiment should preserve total mass, source count, physical regime, construction method, domain, and extraction settings while varying only the source arrangement. Candidate comparison quantities include

- GCS shell volume, area, centroid, and anisotropy distributions;
- retained network face area, seam length, and node inventory;
- low-margin volume fraction and margin distribution;
- fixed-skeleton relabeling fraction;
- tensor-field and complex-fingerprint similarity under declared normalizations.

A successful result would show reproducible arrangement sensitivity beyond bulk source budget. It would not by itself establish observational recoverability or a universal morphology classifier.

5. Legacy proxy evidence

The original three experiments remain in the archival record for four reasons.

First, the proxy rule was explicit, analytically transparent, and easy to falsify. Second, the experiments proved that the extraction pipeline could distinguish hierarchy, density, and arrangement rather than returning only trivial midpoint surfaces. Third, they established support-first discipline: numerical supports and residuals are the evidence objects, while point clouds and meshes are downstream presentations. Fourth, the later decision to withhold superseded dense-scene graphics demonstrates that visual appeal did not outrank extraction integrity.

Their limitations are equally explicit. The proxy fields were assigned source contributions in a weak-field model. They did not include registered comparison between distinct solved geometries, altered nonlinear cross-terms, physical-frame sensitivity, or tensor eigenstructure. Accordingly, the original threshold findings are retained with the qualifier *legacy proxy*:

Table 3: Status of the original GCS IV findings.

Finding	What remains supported	What remains open
Hierarchy persistence	Finite anisotropic support was extractable under the declared proxy.	Registered FN signature and GCS persistence in a fused scene.
Dense emergence	Interface-bearing proxy support could be recovered without imposed tessellation.	Certified FN network topology and persistence.
Morphology sensitivity	Matched budgets produced non-degenerate proxy responses.	Quantified tensor-ablation morphology discrimination.

6. Certification protocol

Promotion from legacy demonstration to certified tensor-ablation result requires a common protocol.

6.1 Scene ledger

Each experiment should report the source roster, masses, positions, physical regime, geometry construction, units, domain, masks, puncture or matter treatment, boundary conditions, and solver tolerances. A Brill–Lindquist time-symmetric vacuum lane is suitable for an exact static benchmark, while extended-density or moving-source lanes must be named separately [4, 5].

6.2 Registration ledger

The point identification, frame map, and reference inner product must be explicit. Shared-coordinate component subtraction may be retained as a declared diagnostic lane, but physical orthonormal-frame comparison remains the preferred public target. Where more than one lawful convention is used, disagreement is a sensitivity result.

6.3 Extraction ledger

The report should include active-support floors, near-tie thresholds, regularization constants, root extraction, topology checks, stratum typing, fixed-skeleton correspondence, and feature-matching tolerances. A visually closed mesh is not sufficient evidence of closure or persistence.

6.4 Sensitivity ledger

At minimum, each principal result should be tested across:

- two or more spatial resolutions;
- Frobenius and spectral norm scalarizations;
- reasonable masks and active-support floors;
- at least one neighboring registration convention where feasible;
- controlled source perturbations;
- render-independent numerical inventories.

6.5 Figure standard

Every principal figure should be regenerated from certified payloads. The preferred panel grammar is:

1. full-scene field or network;
2. ablated-scene field or network;
3. registered $\Delta_i E$ and FN amplitude;
4. extracted GCS, dominance, or margin structure;
5. fixed-skeleton removed, born, and relabeled strata;
6. sensitivity or convergence summary.

Figures are audit windows, not substitutes for the numerical ledger.

6.6 Publication-figure status

The original extreme-ratio point cloud remains a valid illustration of the legacy proxy extraction, but it is not reproduced as principal evidence in this revised paper. Dense-scene and matched-budget graphics from the same lane likewise remain archival artifacts. The principal figures of the certified edition must be regenerated from registered full-versus-ablated payloads and paired with numerical inventories. Until then, the absence of modern result figures is a claim-status safeguard rather than a missing-data disguise.

7. Failure and promotion conditions

The three experiment classes should be weakened or rejected if their structures are dominated by resolution, masking, arbitrary threshold choice, registration convention, or rendering path. A feature that teleports under small source perturbations, vanishes under modest refinement, or appears only in one scalarization without a tensor-level explanation should not be promoted.

Promotion requires more than visual resemblance. A result should show controlled geometric deformation, stable topology or an explicitly diagnosed topology transition, and a quantitative correspondence across the declared analysis family. Morphology sensitivity must exceed within-configuration numerical variability. Dense-scene seams and nodes require residual-gradient independence rather than merely intersecting mesh fragments.

Negative results remain informative. If the subordinate hierarchy signature drowns, if dense networks fail to stabilize, or if matched-budget arrangements become indistinguishable after normalization, the revised framework has identified the limit of the proposed descriptive layer rather than failed to protect a favored image.

8. Relation to GCS I–III

The revised sequence now has a disciplined division of labor:

1. **GCS I** defines the registered source-context parity surface and relational signature.
2. **GCS II** constructs the many-source FN signature network and its cell, face, seam, and node strata.
3. **GCS III** defines margins, fixed-skeleton complex differences, persistence, and relational loss under coarse-graining.
4. **GCS IV** tests whether those objects can be operationally extracted, perturbed, compared, and certified in solved scenes.

GCS IV is therefore not obsolete. It is the experimental gate of the series. The present revision repairs its methods language and evidence hierarchy while leaving the tensor-ablation results explicitly pending.

9. Non-claims

This paper does not report any fused-scene tensor-ablation result for the preserved experimental triad. It does not relabel legacy proxy outputs as tensor-ablation results. It does not modify general relativity, introduce a new force, or define physical membranes, force walls, geodesic tracks, attractors, or universal invariants. It does not claim observational detection or branch-independent uniqueness. It does not claim that a compact parity surface marks the end of a source-conditioned response. It does not claim that a relational cell is sovereign territory.

The narrower claim is that the original experiments established operational recoverability under a declared proxy lane, that their experimental triad remains scientifically well designed, and that the revised GCS framework now supplies a precise certification protocol for repeating them in fused solved scenes.

10. Conclusion

The original GCS IV paper crossed an important threshold: many-source readability geometry could be extracted, not merely imagined. Its three experiments tested persistence under hierarchy,

emergence under complexity, and sensitivity to arrangement. Those results remain valid within the explicit scalar proxy that produced them.

The revised program now asks more. A source-conditioned signature is defined through registered ablation of a solved fused scene, with the electric-Weyl response retained as the primary evidence object and the Frobenius norm used as the default scalar read. Parity surfaces, dominance networks, margins, and relational-complex fingerprints must be extracted from that foundation and survive a declared sensitivity family.

This paper therefore occupies an explicit interim state. The body, terminology, and experimental design are upgraded. The original evidence is preserved at its proper claim level. The next promotion depends on laboratory work rather than prose: rerun the triad, certify the tensor extractions, regenerate the figures, and let the results determine how much of the early morphology survives.

The next task is experimental hardening. The object to be hardened is now the registered tensor-ablation extraction itself.

A. Compact work order for tensor-ablation certification

For each of the three experiment classes:

1. solve the full declared scene;
2. ablate each required source and re-solve;
3. register all comparison tensors to one support;
4. retain $\Delta_i E_{ab}$ and compute $S_i = \|\Delta_i E\|_{F,\text{ref}}$;
5. extract source GCS surfaces, many-source dominance geometry, margins, and fixed-skeleton complex differences as applicable;
6. repeat across the declared sensitivity family;
7. certify numerical inventories before rendering;
8. regenerate publication figures from certified payloads.

References

- [1] D. Scieszinski, “The Gravitational Coherence Surface: Registered Ablation Parity in Relational Gravitational Scenes,” revised formulation, Relational Labs (2026).
- [2] D. Scieszinski, “Many-Source Parity Networks: Registered Ablation Signatures and Relational Gravitational Structure,” revised formulation, Relational Labs (2026).
- [3] D. Scieszinski, “Mesoscale Gravitational Readability: Persistence, Margins, and Relational Complexes Between Local Curvature and Coarse Summary,” revised formulation, Relational Labs (2026).
- [4] D. R. Brill and R. W. Lindquist, “Interaction energy in geometrostatics,” *Physical Review* **131**, 471–476 (1963).
- [5] E.ourgoulhon, “3+1 formalism and bases of numerical relativity,” arXiv:gr-qc/0703035 (2007).
- [6] F. Löffler et al., “The Einstein Toolkit: A community computational infrastructure for relativistic astrophysics,” *Classical and Quantum Gravity* **29**, 115001 (2012).
- [7] D. A. Nichols et al., “Visualizing spacetime curvature via frame-drag vortexes and tidal tendexes. I. General theory and weak-gravity applications,” *Physical Review D* **84**, 124014 (2011).

- [8] T. Buchert, “On average properties of inhomogeneous cosmologies,” *General Relativity and Gravitation* **32**, 105–125 (2000).
- [9] D. Scieszinski, “Proof-of-Concept Extraction of Many-Source Gravitational Readability Structure,” original public theory paper, Relational Labs (2026).